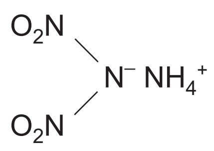
Examiner only

5. The explosive ADN has the structural formula shown below.



Give the empirical formula of ADN. .... [1]

6. Sodium carbonate, Na_2CO_3, is made by heating sodium hydrogencarbonate, $NaHCO_3$.

$$2NaHCO_3 \longrightarrow Na_2CO_3 + CO_2 + H_2O$$

M_r 84 $\qquad$ M_r 106

Calculate the atom economy of this reaction. [1]

Atom economy = ... %

2410U101
03

7. Hydrogen and iodine react together to give hydrogen iodide.

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

(a) Write the expression for the equilibrium constant in terms of concentration, K_c, for this reaction. [1]

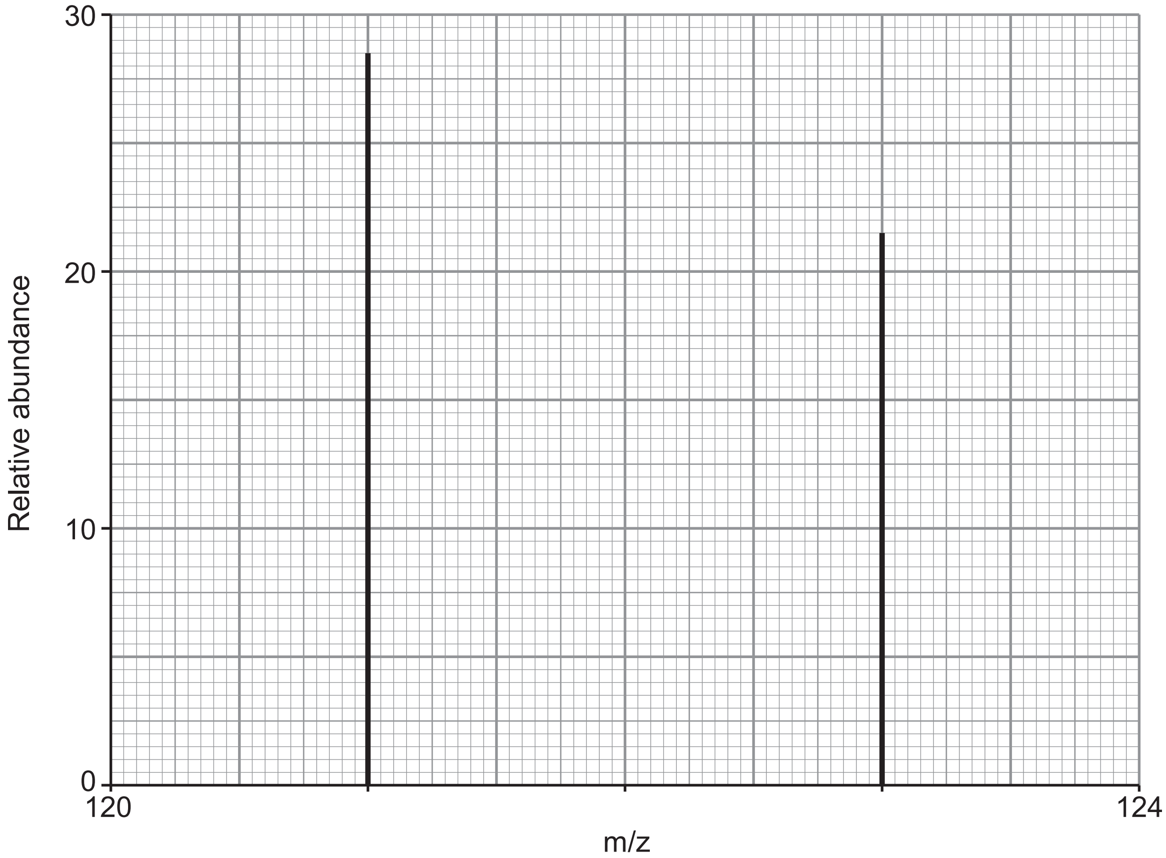
(b) The equilibrium constant, K_c, for this reaction has a value of 46.0 at a certain temperature. Calculate the equilibrium concentration of hydrogen at this temperature. The equilibrium concentration of iodine is 1.20 mol dm^{-3} and the equilibrium concentration of hydrogen iodide is 15.00 mol dm^{-3}. [2]

Concentration of hydrogen = ... mol dm^{-3}

10

Examiner only

(c) Antimony is in Group 5. Its mass spectrum shows that it has two stable isotopes.



Calculate the relative atomic mass of antimony. You **must** show your working. [2]

Relative atomic mass = ...

(d) Although radiation from radioisotopes is harmful to health many beneficial uses of radioactivity have been found.

The table below gives some information about four radioactive isotopes.

Isotope	Radiation emitted	Half-life
^{90}Sr	β	28 years
^{99}Tc	γ	6 hours
^{210}At	α	8.1 hours
^{228}Th	α	1.9 years

Use **all** the data given to choose which isotope is the most suitable to use as a tracer in medicine. Explain your answer. [3]

....

....

....

....

....

....

12

Examiner only

5. Give the oxidation number of vanadium in $VOCl_3$. [1]

....

6. Phosphoric acid has the formula H_3PO_4. Write the formula of magnesium phosphate. [1]

....

7. Cooking fuel for outdoor camping contains butane, which reacts with oxygen according to the following equation.

$$C_4H_{10}(g) \quad + \quad 6\tfrac{1}{2}\,O_2(g) \quad \longrightarrow \quad 4CO_2(g) \quad + \quad 5H_2O(g)$$

If 1.00 mol of butane reacts in this way, calculate the number of **molecules** of carbon dioxide that will be formed. [1]

Molecules of CO_2 = ...

8. 9.60 g of titanium reacts completely with 3.68 dm^3 of oxygen gas at 298 K and 1 atm to form an oxide.

Calculate the empirical formula of this oxide. [2]

Empirical formula ....

10

Examiner only

5. The ammonium ion contains a 'coordinate bond'. Explain what is meant by this term. [1]

....

....

6. Uranium is used in nuclear fuel reactors. One of its isotopes, uranium-235, has a half-life of 7.03×10^8 years and decays by α-emission.

(a) Give the mass number and symbol of the element formed as a product of the radioactive decay of uranium-235. [1]

....

(b) If a quantity of uranium-235 decays, state what fraction is left after 2.812×10^9 years. [1]

fraction left = ...

2410U101
03

7. Calculate the mass of calcium that contains the same number of atoms as there are molecules in 9.1 g of sulfur dioxide, SO_2. [2]

calcium mass = ... g

10